

FLORENCE BOCKTING

Postdoctoral Researcher & Research Software Engineer

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Profile

Research scientist and research software engineer specialising in Bayesian methods, computational modelling, and reproducible research software. Currently a postdoc and research software engineer in the Bayesian Workflow Group at Aalto University, implementing Bayesian methods in open-source, well-tested software within the Stan ecosystem. Doctorate in computational statistics from TU Dortmund University, with research on simulation-based expert prior elicitation and authorship of the Python package `elicito`, distributed via PyPI and conda-forge. Background spanning psychology, cognitive science, and climate science, with experience contributing to open-source projects in Python, R, and C++.

Core Expertise

- Bayesian statistics
- Computational modelling
- Machine learning
- Research software engineering
- Open-source scientific software
- Reproducible and open science
- Data pipelines
- Cognitive, social, and climate science

Programming & Tools

Languages	Python, R, Stan; basic knowledge of C++ and OCaml
Version control & CI	Git, GitHub, GitLab, GitHub Actions
Testing & code quality	`pytest`, `testthat`, `ruff`, `black`, `mypy`, `lintr`
Packaging & environments	`uv`, `pip`, `poetry`, `pixi`, `renv`, PyPI, conda-forge, CRAN
Documentation	Sphinx, Read the Docs, `roxygen2`, R vignettes
Writing & reporting	Quarto, Jupyter, Jupytext, R Markdown, LaTeX, Overleaf, Shiny
Pipelines & compute	Prefect, `targets`, HPC (LIDO3, NERSC), Slurm clusters
Open science & data	Zenodo, OSF
Environments	Linux / Ubuntu, Windows
Lifecycle management	Reverse dependency checks, migration guides, deprecation management

Further Skills

Languages	German (native), English (fluent), French (intermediate), Finnish (beginner)
Project management	Jira, Confluence, Notion
Communication	Slack, Zulip, Zoom
Writing & references	Zotero, Microsoft Office, LibreOffice / OpenOffice
Graphics	GIMP

Experience

Postdoctoral Researcher / Research Software Engineer 2026–present
Bayesian Workflow Group, Department of Computer Science, Aalto University
Espoo, Finland · PI: Prof. Dr. Aki Vehtari

- Implement Bayesian statistical methods as tested, documented, open-source software, primarily within the Stan ecosystem.
- Develop and maintain R packages by translating current Bayesian workflow research into robust software and improving existing package architecture, tests, and documentation.

Research Scientist / Research Software Engineer 2025–2026
Climate Resource GmbH
Berlin, Germany

- Developed greenhouse gas concentration data pipelines for earth-system modelling groups.
- Designed and implemented an ESA-funded pipeline for scraping, preprocessing, and assimilating multi-source observational data from ground-based and satellite sources, resulting in a unified, standardised data product.

Consultant Aug–Dec 2025
Jacobs Foundation
Zürich, Switzerland

- Implemented expert prior elicitation to support internal decision-making processes.

Research Scientist — 2023–2025

Computational Statistics, Department of Statistics, TU Dortmund University
Dortmund, Germany · PI: Prof. Dr. Paul-Christian Bürkner

SC SimTech, University of Stuttgart — 2022–2023

Stuttgart, Germany · PI: Dr. Paul-Christian Bürkner

- Conducted doctoral research in computational statistics on simulation-based expert prior elicitation.
- Developed a method for learning prior distributions in Bayesian models from expert knowledge.
- Implemented the approach as the open-source Python package `elicito`, distributed via PyPI and conda-forge.

Research Scientist — 2020–2022

Methods & Statistics, Department of Psychology, Philipps University of Marburg
Marburg, Germany · PI: Prof. Dr. Daniel W. Heck

- Worked on statistical method development, computational and mathematical modelling, social psychology, and the formalisation of verbal theories.
- Formalised a verbal theory of truth judgments and the truth effect in social psychology by developing a computational model in R.

Tutor & Student Assistant — 2015–2020

Harz University of Applied Sciences; University of Osnabrück
Wernigerode and Osnabrück, Germany

- Supported lecture preparation and led exercise seminars across courses including Data Ethics, Statistics, Computational Data Analysis, Bayesian Data Analysis, General Psychology, and Experimental Psychology.

Research Assistant — 2018–2019

Quantitative and Qualitative Healthcare, Produkt+Markt GmbH
Osnabrück, Germany

- Conducted qualitative and quantitative market research in the healthcare sector.
- Coded qualitative questionnaires, performed descriptive data analyses, and prepared presentations and reports.

Project Member — Student Skills Matching Platform — 2018

Harz University of Applied Sciences
Wernigerode, Germany

- Contributed to the development of a platform for matching students based on skills.
- Carried out a needs analysis and led the conceptual design of the matching system.

Internships

- **Qualitative Market Research in Healthcare**, Ipsos GmbH, Hamburg, Germany, 2017
- **General Psychology and Methodology**, University of Bamberg, Germany, PI: Prof. Dr. Claus-Christian Carbon, 2016

Selected Software and Open-Source Contributions

Maintainer / Author

- **`elicito`** (Python) — Expert prior elicitation: learning prior distributions of model parameters from expert knowledge. Distributed via PyPI and conda-forge, documented on Read the Docs.
- **`gradient-aware-harmonisation`** (Python) — Harmonisation of two time series using zero- and first-order gradient information.
- **`GHG-forcing-for-CMIP`** (Python) — Pipeline constructing a composite greenhouse gas dataset from ground-based and satellite observations, suitable as forcing input for CMIP climate models.

Contributor

- **Stan ecosystem** — `loo` (leave-one-out cross-validation), `bayesplot` (posterior and MCMC visualisation), `posterior` (draws manipulation), `brms` (Bayesian regression interface), and `stan-dev/math` (C++ autodiff and distributions): feature development, refactoring, testing, documentation, reverse dependency checks, and migration guides.
- **CMIP7 climate data** — `cmip7-scenariomip-ghg-concentrations` (ScenarioMIP forcing generation), `CMIP7-GHG-Concentration-Manuscripts` (reproducible Jupytext-to-LaTeX manuscript workflow), and `input4MIPs_CVs` (controlled vocabularies for input4MIPs metadata).
- **RSE survey analysis tooling** — Analysis of results from the International RSE Survey for the Nordic RSE community and development of a reproducible analysis tool for reuse across RSE communities.
- **Open-source community work** — Vignettes, tutorials, pull request reviews, issue triage, and mentoring.

Education

Doctorate in Computational Statistics — 2026
TU Dortmund University
Dortmund, Germany
Thesis *Simulation-based expert prior elicitation: Method and software development*
- Doctoral thesis conducted at SimTech / University of Stuttgart and TU Dortmund University.
- Awarded *magna cum laude*.

M.Sc. in Cognitive Science — 2020
University of Osnabrück
Osnabrück, Germany
- Majors: Cognitive modelling and artificial intelligence.
- Graduated with distinction.

B.Sc. in Business Psychology — 2018
Harz University of Applied Sciences
Wernigerode, Germany
- Majors: Market research and consumer behaviour.
- Graduated with distinction.

Marketing Communications Specialist — 2014
Dresden-Informatik GmbH
Dresden, Germany
- Vocational training certified by IHK, Chamber of Industry and Commerce, Germany.

Teaching & Thesis Supervision

Master Seminar on Multilevel Modelling — 2023–2024
- Target audience: Master's students in Data Science, Statistics, and Econometrics.
- Introduced the theory and analysis of multilevel models using R from both Bayesian and frequentist perspectives, implemented with `lme4` and `brms`.
- Language of instruction: English.

Programming Course: Introduction to Python — 2023–2025
- Target audience: Bachelor's and Master's students in Data Science, Statistics, and Econometrics.
- Covered Python fundamentals, data wrangling and visualisation, documentation with Sphinx, testing with `pytest`, and version control with Git and GitHub; course material published as a public repository.
- Languages of instruction: English and German.

Supervision of Theses — 2020–2025
- B.Sc. thesis: *Analysis of different initialization approaches for hyperparameter optimization with mini-batch stochastic gradient descent: A simulation study* · TU Dortmund University
- M.Sc. thesis: *Sensitivity analysis and performance evaluation of varying upper thresholds for discrete likelihoods using the Softmax-Gumbel Trick* · TU Dortmund University
- B.Sc. thesis: *The influence of response scales on the knowledge gain of underlying cognitive mechanisms: The role of uncertainty and truth perception in the Truth Effect* · Philipps University of Marburg
- B.Sc. thesis: *Empirical test of core assumptions of the Referential Theory: Influence of repetition on perceived coherence* · Philipps University of Marburg
- B.Sc. thesis: *Identification and testing of relevant psychological factors on truth judgments and the truth effect according to the Referential Theory* · Philipps University of Marburg
- B.Sc. thesis: *Truth Effect: The role of the response scale in truth effect designs with short delay* · Philipps University of Marburg

Publications & Talks

- Bockting, F., Bronder, S., Vehtari, A., & Ward, B. *Contributing to Stan: A Developer's Guide to the Core, Interfaces, and First Contributions.* Tutorial at StanCon26, Uppsala, Sweden. — 2026
- Bockting, F. *Simulation-based expert prior elicitation: Method and software development.* Doctoral thesis, TU Dortmund University.
- Bockting, F. *Predictive Prior Elicitation: State of the Art & Current Challenges.* Contributed talk, Workshop on Bayesian Modelling, Lund University. — 2025
- Bockting, F., & Bürkner, P. C. *elicito: A Python Package for Expert Prior Elicitation.* arXiv preprint.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Expert-elicitation method for nonparametric joint priors using normalizing flows. Statistics and Computing*, 35(5), 132.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Normalizing Flows for Simulation-Based Expert Prior Elicitation.* Contributed talk, MathPsych. — 2024

- Bockting, F., Radev, S. T., & Bürkner, P. C. *Simulation-Based Prior Knowledge Elicitation for Parametric Bayesian Models.* Contributed talk, ISBA.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Simulation-Based Prior Knowledge Elicitation for Parametric Bayesian Models.* Invited talk, Bayes@Lund.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Simulation-based prior knowledge elicitation for parametric Bayesian models. Scientific Reports*, 14, 17330.
- Heck, D. W., & Bockting, F. *Benefits of Bayesian model averaging for mixed-effects modeling. Computational Brain & Behavior*, 6(1), 35–49. 2023
- van Doorn, J., ..., Bockting, F., & Aust, F. *Bayes factors for mixed models: A discussion. Computational Brain & Behavior*, 6(1), 1–13.
- Bockting, F., & Heck, D. W. *Measuring Individual Differences in the Truth Effect: A Formal Analysis.* Fast talk at MathPsych. 2021
- Stephan, A., ..., Bockting, F., ... *Nachwort.* In A. M. Turing, *Computing Machinery and Intelligence. Können Maschinen Denken?* Reclam, pp. 131–201.

Workshops & Courses

- Developing Secure Software, OpenSSF, Linux Foundation 2026
- High Performance R, CSC – IT Center for Science, Finland
- Copyright for Computer Programs & Software, TU Dortmund University 2024
- Research Software Engineering Summer School, Karlsruhe Institute of Technology, KIT
- Theory of Science, Prof. Dr. Zoglauer, University of Stuttgart 2023
- Foundations of Deep Learning for the Social Sciences, University of Tübingen
- The Statistics Wars and Their Casualties, online seminar series, Prof. Dr. Deborah Mayo, Prof. Dr. Roman Frigg, and Prof. Dr. Margherita Harris
- Summer School on Advanced Bayesian Data Analysis with Stan, Dr. Bruno Nicenboim, University of Potsdam 2022
- Interval Hypothesis Testing, Prof. Dr. Daniël Lakens, Eindhoven University of Technology
- Robust Cognitive Bayesian Analysis, Prof. Dr. Jeffrey N. Rouder, University of California
- Bayesian Evaluation of Informative Hypotheses, Prof. Dr. Herbert Hoijtink, Utrecht University
- Multinomial-Processing-Tree Modeling: Foundations and Recent Advances, Prof. Dr. Edgar Erdfelder and Prof. Dr. Daniel Heck, University of Mannheim 2021
- Single- vs. Dual-Process Theories, Prof. Dr. Mandy Hütter, University of Tübingen
- Introduction to Bayesian Statistics, Prof. Dr. Daniel Heck, Philipps University of Marburg

Conferences & Events

- StanCon — Stan Community, Uppsala
- MathPsych — Society for Mathematical Psychology, Tilburg
- ISBA — International Society for Bayesian Analysis, Venice
- DGP — German Psychological Society, Leipzig
- EGU — European Geosciences Union, Vienna
- Bayes@Lund, Lund

Engagement

- Analysed results from the International RSE Survey for the Nordic RSE community and developed a reproducible analysis tool for reuse across RSE communities worldwide.
- Contributed to the HiRSE Summary of AI for RSE with a video on “AI-assisted contributions in Open Source”.
- Carpentries lesson maintainer for “HPC Workflow Management with Snakemake”.
- Mentor in Google Summer of Code 2026 for “Parallelizing `loo` package”.
- Co-instructor for the StanCon 2026 tutorial “Contributing to Stan: A Developer’s Guide to the Core, Interfaces, and First Contributions”.
- Author of “Contributing to Stan”, onboarding material guiding new contributors through their first contribution to the Stan ecosystem.

Awards & Volunteering

- Fellow of the Studienstiftung des deutschen Volkes (German Academic Scholarship Foundation), 2015–2020.
- Volunteer at KANA Dortmund (soup kitchen), Dortmund, Germany.
- Volunteer at a residential home for children and adolescents with disabilities, Lebenshilfe e.V.